Production Matching Coverage & Engineering Gap Analysis (V2)

To: ISKONNECT Stakeholders

From: ISKONNECT Team of Experts

Objective: Evaluation of engineering readiness to faithfully implement canonically verified scholarship rules.

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1. Executive Summary: Production Readiness Assessment

Based on a synthesis of the **Canonical Verification Reports** (Group A, B, and C) against the **Current Database Schema** and **Matching Engine** documentation, the platform **cannot** currently faithfully represent every officially verified scholarship rule without producing incorrect recommendations.

While the engine handles basic GWA and income filters, it lacks the **structural depth** to model specialized entry gates (e.g., zero-unit clauses), complex sectoral registries (e.g., NCFRS), and institutional exclusivity mandates (e.g., LGU conflicts). **Production risk is currently High** due to data corruptions and schema omissions that will trigger significant false-positive recommendations.

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2. Key Engineering Gaps & Production Risks

A. Temporal and Entry-Gate Logic (High Risk)

The current system fails to distinguish between generic year-level eligibility and strict entry-window constraints.

- **Gap:** The schema lacks an `incoming_year_only` or `max_earned_units` boolean.
- **Rule Conflict:** Programs like **DOST Undergraduate (ID 73)** and **CHED BPMSP (ID 76)** strictly require "zero (0) earned tertiary units." Conversely, **Aboitiz Brights (ID 75)** is restricted *exclusively* to Year 2 students.
- **Production Impact:** Enrolled upperclassmen will incorrectly match with freshman-only grants, leading to automatic disqualification by providers.

B. Sectoral and Affiliation Hard-Filters (Critical Risk)

The engine relies on generic `priority_groups` (JSONB) intended for scoring, rather than deterministic hard-gates for sectoral membership.

- **Gap:** Missing explicit fields for **Parent Employment** and **Sectoral Registry Status**.
- **Rule Conflict:**
  - **CoScho (ID 117):** Requires parent registration in the **NCFRS** coconut farmer database.
  - **DA ACEF (ID 55):** Requires parent registration in the **RSBSA**.
  - **OWWA ELAP (ID 124):** Requires parent status to be specifically "deceased" or "permanently incapacitated."

- **Production Impact:** General agriculture students or active OFW families will receive false-positive matches for programs requiring specific casualty or registry status.

C. Academic and Quantitative Merit Alternatives

The current schema and engine treat `min_gwa` as the primary academic gate, failing to model "either/or" merit logic.

- **Gap:** Missing a `rank_cutoff_alternative` field.
- **Rule Conflict: CHED BPMSP (ID 76)** allows entry if the student has a 95% GWA **OR** is in the Top 5 of their class. **Makati (ID 31, 98)** and **PUP (ID 68)** use similar Top 10% class rank triggers.
- **Production Impact:** Top-ranked students with slightly lower GWAs (e.g., 93% GWA but Rank 2) will be incorrectly filtered out (false-negatives).

D. Institutional Exclusivity & Conflict Resolution

The platform does not currently model the "One-Grant-Only" policies mandated by LGUs and national agencies.

- **Gap:** No `lgu_exclusivity_clause` or `national_grant_exclusivity` fields.
- **Rule Conflict: Quezon City (ID 88)** and **Taguig (ID 27-29)** explicitly bar students already enjoying another LGU grant. Most CHED/DOST programs bar duplicate national funding.
- **Production Impact:** Students already holding a grant will be matched with incompatible secondary funding, causing administrative friction.

E. Professional and Graduate Tracks

The current schema cannot handle professional prerequisites for international or graduate tracks.

- **Gap:** Missing `work_experience_years` and `age_limit` constraints.
- **Rule Conflict: Australia Awards (ID 74)** and **Chevening (ID 91)** require a minimum of 2 years post-graduation work experience. **MEXT (ID 81-83)** and **GKS (ID 65)** enforce strict age cutoffs based on specific birthdates (e.g., April 2, 2002).

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3. Strategic Data Quality Anomalies

The current database contains high-risk "silent failures" where the engineering state contradicts verified policy:

- **ID 130 (JLSS):** Marked `is_active: false` (Archived) while verified as an active, mandatory annual program for 2nd-year students.
- **ID 3 (DOST Graduate):** Erroneously enforces a **₱500,000 income ceiling** and underestimates stipends by ~50%.

- **ID 54 (CHED MSRS):** Records an income cap of ₱600k in the DB, while verified CMO rules set it at **₱450k**, creating a ₱150k window of false-positive eligibility.

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4. Required Engineering Actions (Directives)

I. Schema Expansion (DDL)

Database Architects must execute the following structural updates to `public.scholarships`:

- `ADD COLUMN incoming_year_only BOOLEAN DEFAULT true` (Freshman entry gate).
- `ADD COLUMN alternative_class_rank INT` (Top 5, Top 10 logic).
- `ADD COLUMN sectoral_restriction VARCHAR` (NCFRS, RSBSA, SRA tags).
- `ADD COLUMN lgu_exclusivity_clause BOOLEAN` (LGU conflict filter).
- `ADD COLUMN work_experience_years INT` (International/Professional gate).
- `ADD COLUMN partner_school_restricted BOOLEAN` (Consortium/Partner school lock).

II. Matching Engine Pipeline Enhancement

Backend Engineers must implement a **Three-Stage Pipeline** in `match_service.py`:

1. **Stage 1 (Hard Temporal/Level):** Enforce `earned_units == 0` for freshman-only tracks and dynamic age calculation against birthdate.
2. **Stage 2 (Quant/Socioeconomic):** Implement `OR` logic for `(gwa >= min_gwa OR class_rank <= rank_cutoff)`.
3. **Stage 3 (Sectoral/Institutional):** Cross-verify user affiliation (e.g., `user.is_ncfrs_registered`) and `school_id` against specific partner/consortium lists.

III. Data Remediation (DML)

All Group B/C programs identified as "Published" or "Open" must undergo a batch update to align with Tier 1 evidence (e.g., Correcting ID 3, 54, 130, and 117).

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5. Final Conclusion

The ISKONNECT platform is **not production-ready** for its deterministic matching mission. While the verification reports provide the necessary "intelligence," the **current schema is too shallow** to house that intelligence. Engineering work must shift from UI/UX and ranking to these **fundamental structural extensions** to prevent misleading Filipino students.

ISKONNECT Production Matching Coverage & Engineering Gap Analysis (V2)

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1. Executive Summary

Based on a synthesis of the **Live Database Export** and **Canonical Verification Reports (V1–V3)**, the following metrics define the current state of the ISKONNECT platform:

- **Total Scholarships Reviewed:** 120.
- **Active Scholarships:** 114.
- **Archived Scholarships:** 6.
- **Scholarships Requiring Manual Review:** 83 (Editorial state: `needs_review`).
- **Overall Engineering Readiness: LOW.** While the engine processes basic academic and geographic filters, it cannot currently enforce the "Zero-Unit" clauses, "One-Scholar-per-Family" rules, or sectoral registry verifications required by the verified rules.
- **Overall Schema Coverage Estimate: ~65%.** The current schema handles approximately two-thirds of required parameters but lacks the structural depth for temporal logic, professional prerequisites, and institutional exclusivity.
- **Overall Production Risk: CRITICAL.** There is a high probability of **false-positive recommendations** for freshmen-only grants and **false-negative omissions** for top-ranked students with lower GWAs due to missing "OR" logic (GWA vs. Class Rank).

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2. Eligibility Rule Inventory

A master inventory of rules discovered across the verification audits:

Rule Type	Importance	Existing Support	Missing Support / Gaps
Academic Standing (GWA)	Critical	`min_gwa_normalized`	Needs "Either/Or" logic for Rank alternatives.
Enrollment Timing	High	`application_status`	`max_earned_units` and `incoming_year_only`.
Sectoral Registry	High	`priority_groups` (JSONB)	Hard-gates for **RSBSA, NCFRS, and SRA**.

Grant Exclusivity	High	None	`lgu_exclusivity` and `national_grant_bar`.
Household Limit	High	None	`one_scholar_per_family` cross-field check.
Professional Exp.	Medium	None	`work_experience_years` for Int'l grants.
Institution Lock	Critical	`eligible_schools`	**Consortium-only** table lookups (ERDT/ASTHRDP).
Age Cutoffs	High	`min_age / max_age`	Dynamic logic for **birthdate-specific** cutoffs.
Return Service	Medium	`has_return_service`	**Service Type** (Teaching vs. Health vs. Industry).

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3. Schema Coverage Assessment

Eligibility Rule	Used By	Existing Schema	Partial Support	Missing	Engineering Recommendation
Zero-Unit Clause	DOST (73), BPMSP (76)			[X]	ADD `incoming_year_only` BOOLEAN.

Class Rank Alt.	Makati (31), BPMSP (76)	[X]		ADD `rank_cutoff_alternative` INT.
LGU Exclusivity	QC (88), Taguig (27)		[X]	ADD `lgu_exclusivity_clause` BOOLEAN.
Sectoral Registry	CoScho (117), DA (55)	[X]		ADD `sectoral_restriction` ENUM.
Family Cap	Pasig (25), Davao (37)		[X]	ADD `one_scholar_per_family` BOOLEAN.
Work Experience	Chevening (91), Australia (74)		[X]	ADD `work_experience_years` INT.
Employment Status	GSIS (78), SIKAP (120)	[X]		ADD `parent_salary_grade` & `is_hei_staff`.

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4. Matching Logic Coverage

- **Rules Already Enforced:** Basic GWA cutoffs, Region/City residency, Income ceilings (though many caps are currently corrupted in data), and School types.
- **Rules Partially Enforced:** `priority_groups` as soft scores rather than hard gates, and `eligible_levels` which lack lateral entry distinction.
- **Rules Not Enforceable:**
  - **Compound/OR Conditions:** e.g., "GWA >= 95 **OR** Top 5 Rank".

    - **Historical State:** "Has **NOT** earned tertiary units" requires checking the student's completed unit count.
    - **Cross-Field Validation:** "One scholar per family" requires querying existing grants held by family members.
    - **Temporal Logic:** Birthdate-calculated age relative to intake date (e.g., "Born after April 2, 1992").
- **Rules Requiring Normalization:**
    - **GWA Scales:** Normalizing university-specific decimal scales (1.0–5.0) against percentage-based scholarship rules.
    - **Registry Tags:** Mapping user-provided employment proof to agency registries like RSBSA or NCFRS.

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5. Hidden Eligibility Risks

1. **The "Enrolled Freshman" Trap (Zero Units):**
    - **Risk:** Enrolled college freshmen will match with DOST (73) or BPMSP (76).
    - **Reason:** These grants require **zero (0)** post-secondary units. Once a student finishes their first semester, they are permanently barred from these "Freshman Track" grants. The current engine lacks an `earned_units == 0` check.
2. **LGU and National Grant Exclusivity:**
    - **Risk:** Students holding a Taguig grant (27) will match with Quezon City (88) or CHED Merit (1).
    - **Reason:** Most LGUs and national agencies strictly bar dual-grant enjoyment. Recommending a second grant leads to immediate disqualification or refund demands.
3. **Consortium Institutional Locks:**
    - **Risk:** Students in any public university will match with DOST Graduate (3) or ERDT (134).
    - **Reason:** These are restricted to **Consortium Member Universities** (only 8-10 specific schools). Generic "SUC" filters will produce massive false positives.
4. **Implicit Marital/Health Status:**
    - **Risk:** Married students matching with OWWA (85) or Caritas (59).
    - **Reason:** These programs explicitly mandate **Single Marital Status**, a field currently under-utilized in the hard-filter layer.
5. **Graduate vs. Undergraduate Med/Law:**
    - **Risk:** Pre-med seniors matching with CHED MSRS (54).

- **Reason:** The program requires a **completed Bachelor's degree and NMAT score** *prior* to entry. The engine must distinguish "Incoming Graduate" from "Ongoing Undergraduate".

Production Matching Coverage & Engineering Gap Analysis (V2)

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1. Executive Summary

A comprehensive audit of the ISKONNECT platform against canonical verification reports (V1–V3) reveals a significant divergence between officially verified scholarship rules and the platform's current engineering capabilities.

- **Total Scholarships Reviewed:** 120.
- **Active Scholarships:** 114.
- **Archived Scholarships:** 6.
- **Scholarships Requiring Manual Review:** 83 (currently in `needs_review` editorial state).
- **Overall Engineering Readiness: LOW.** The system lacks the structural depth to handle strict temporal gates (Zero-Unit clauses), sectoral registry verifications (RSBSA/NCFRS), and institutional exclusivity mandates (LGU conflicts).
- **Overall Schema Coverage Estimate: ~65%.** While basic academic and demographic fields exist, secondary control parameters (e.g., `incoming_year_only`, `lgu_exclusivity`) are missing from the `public.scholarships` table.
- **Overall Production Risk: CRITICAL.** Without immediate schema mutations and matching engine logic updates, the platform will generate significant false-positive recommendations for freshman-only and sectoral grants.

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2. False Positive Analysis

Missing Capability	Current Behavior	Incorrect Recommendation	Affected Scholarships	Production Impact	Severity

Zero-Unit Clause	Filters by `year_level` only.	Recommends freshman grants to enrolled college students.	DOST (73), BPMSP (76), SM (10).	Students waste effort applying for grants they are permanently barred from after earning 1 unit.	**Critical**
LGU Exclusivity	Checks only residency.	Recommends QCSP (88) to a student already holding a Taguig grant.	QC (88), Taguig (27-30).	Administrative friction; dual-enjoyment leads to disqualification or refund demands.	**High**
Sectoral Registry	Checks only degree type.	Recommends agriculture grants to non-farming families.	CoScho (117), DA (55), SIDA (118).	Massive rejection rates; these programs require SRA or NCFRS registry proof.	**High**
Institutional Lock	Checks only `school_type`.	Recommends ERDT (134) to students in non-consortium SUCs.	ERDT, ASTHRDP, Aboitiz (75).	Misleading students; these grants are restricted to ~8–10 specific universities.	**Critical**

3. False Negative Analysis

- **Logic Limitation:** Missing **"Either/Or" (OR) logic** for academic cutoffs.
- **Affected Scholarships:** BPMSP (76), Makati (31).

- **Impact:** A student with a 93% GWA who is **Top 5 in their class** will be rejected by the current engine because it only evaluates `min_gwa: 95`. This suppresses valid funding for the most elite candidates.

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4. Database Engineering Gap Analysis (Grouped Issue Matrix)

A. Temporal and Enrollment Gaps

- **Current Limitation:** Lacks fields for "Zero units earned" or "Lateral entry allowed".
- **Affected Fields:** `eligible_year_levels`, `incoming_year_only` (Missing).
- **Affected Scholarships:** DOST (73, 130), Aboitiz (75), SM Foundation (10).
- **Recommendation:** `ADD COLUMN incoming_year_only BOOLEAN`, `ADD COLUMN max_earned_units INT`.

B. Sectoral and Affiliation Gaps

- **Current Limitation:** Relies on generic `priority_groups` (JSONB) for soft-scoring instead of hard deterministic gates.
- **Affected Fields:** `sectoral_restriction`, `parent_employment_restriction`.
- **Affected Scholarships:** CoScho (117), OWWA (85, 124), GSIS (78, 84).
- **Recommendation:** `ADD COLUMN sectoral_restriction VARCHAR`, `ADD COLUMN parent_gsis_sg INT`.

C. Institutional Gaps

- **Current Limitation:** No lookup table for Consortium/Partner University locks.
- **Affected Fields:** `partner_school_restricted` (Missing).
- **Affected Scholarships:** Megaworld (61), ERDT (134), ASTHRDP (133).
- **Recommendation:** `ADD COLUMN partner_school_restricted BOOLEAN`, link to `organizations` table.

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5. Matching Engine Gap Analysis

Requirement Type	Why it is Required	Affected Scholarships	Complexity	Regression Risk
OR Logic	For "GWA >= X **OR** Rank <= Y" rules.	BPMSP (76), Makati (31).	Medium	Low

Cross-Field Validation	For "One scholar per family" rules.	Pasig (25), Davao (37).	High	Medium
Temporal Age Logic	Age must be relative to intake date, not today's date.	MEXT (81-83), GKS (65).	Medium	Medium
Salary Grade Logic	Parent income ceiling is tied to civil service SG levels.	GSIS GSSP (78).	Low	Low

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6. Engineering Priority Matrix

- **P0 — Critical (Immediate Fix Required):**
  - Implementing the **"Zero-Unit" clause** to prevent enrolled freshmen from matching with DOST/BPMSP.
  - Fixing the **LGU exclusivity gate** to prevent duplicate local grant recommendations.
  - Correcting **corrupted data** for ID 3 (DOST Graduate) which currently enforces a false ₱500k cap.
- **P1 — High:**
  - Adding **Consortium Lookup** validation for institutional locks.
  - Implementing **OR logic** for GWA vs. Class Rank.
- **P2 — Medium:**
  - Normalization of GWA scales (decimal vs. percentage) for deterministic matching.

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7. Production Readiness Assessment

- **Overall Schema Coverage: 65%** [Source Synthesis].
- **Overall Matching Coverage: 50%** (Fails on all complex logic types) [Source Synthesis].
- **Critical Blockers:** Missing `earned_units` user field; Missing `exclusivity_clause` scholarship field.

- **High-Risk Assumption:** Currently assumes `eligible_levels` implies freshman entry; verified reports prove this is false.

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8. ISKONNECT Engineering Specification for Production Matching

I. Engineering Principles

1. **Hard-Gate First:** Quantitative filters (GWA, Income, Units) must execute before soft scoring.
2. **Zero-False-Positives:** Recommending a scholarship to a candidate with 1 earned college unit when the grant requires 0 units is a system failure.
3. **Temporal Relativity:** Eligibility is calculated against the **Academic Year Target**, not the current system time.

II. Required Schema Changes (DDL)

None

```
ALTER TABLE public.scholarships
  ADD COLUMN IF NOT EXISTS incoming_year_only BOOLEAN DEFAULT false,
  ADD COLUMN IF NOT EXISTS max_earned_units INT DEFAULT NULL,
  ADD COLUMN IF NOT EXISTS rank_cutoff_alternative INT DEFAULT NULL,
  ADD COLUMN IF NOT EXISTS lgu_exclusivity_clause BOOLEAN DEFAULT
false,
  ADD COLUMN IF NOT EXISTS one_scholar_per_family_clause BOOLEAN
DEFAULT false,
  ADD COLUMN IF NOT EXISTS sectoral_restriction VARCHAR(100) DEFAULT
NULL,
  ADD COLUMN IF NOT EXISTS partner_school_restricted BOOLEAN DEFAULT
false,
  ADD COLUMN IF NOT EXISTS work_experience_years INT DEFAULT NULL;
```

(Ref:)

III. Required Matching Logic Changes

- **Stage 1 Update:** If `incoming_year_only == true`, engine **must** verify `user.earned_college_units == 0`.
- **Stage 2 Update:** Implement `IF (GWA >= min_gwa OR class_rank <= rank_cutoff_alternative) THEN qualified`.
- **Stage 3 Update:** If `lgu_exclusivity_clause == true`, engine **must** check `user.existing_grants` for other LGU tags.

IV. Required Explainability Changes

- **Temporal Warnings:** "You will become eligible for this grant in your Junior Year" (JLSS ID 130).

- **Disqualification Alerts:** "You are ineligible for this grant because you have already earned college units" (DOST ID 73).

V. Regression Testing Requirements

- **Persona A (The "Enrolled Freshman"):** Verify that a student in 1st Year, 2nd Sem with 18 units **cannot** match with DOST (73).
- **Persona B (The "Elite Ranker"):** Verify that a student with 90% GWA (below 95% threshold) but **Rank 2 in class can** match with BPMSP (76).
- **Persona C (The "Double Dipper"):** Verify that a student holding a Taguig grant is filtered out of QCSP (88).

VI. Success Criteria for Production Readiness

- 100% of Group B/C programs have the `incoming_year_only` flag correctly set.
- The `max_income` parameter for ID 3 is removed (uncapped per policy).
- The engine correctly handles "Consortium Only" matching for ERDT and ASTHRDP.